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Anaphylaxis and Allergic Reactions

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Session Objectives

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- Discuss the four hypersensitivity reactions, with the main emphasis on Type 1 hypersensitivity reactions.

- Identify the clinical manifestations of allergies & the differential diagnosis of allergic reactions

- Recognize when a patient is having an anaphylactic reaction

- Treatment of anaphylaxis and non-anaphylactic allergic reactions

Definitions

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Antigen (Ag)– foreign or self-molecules that trigger immune responses

Antibody (Ab)– B cell produced molecules encoded by genes that have heavy and light chains that bind to Antigens

Ex: IgG, IgE, IgM

Terms

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- There is a normal protective role of IgE

In host defense.

- Allergy – abnormal IgE mediated

Immune response to an otherwise harmless substance (antigen)

- Anaphylaxis – antithesis of prophylaxis which means ”toward protection.”

Terms

- Complement – plasma proteins that play a role in innate immunity and inflammation

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- **Hypersensitivity Reactions**
- Immunological responses that affect the host adversely
- I-IV

Warren Levinson, Peter Chin-Hong, Elizabeth A. Joyce, Jesse Nussbaum, Brian Schwartz

Listen

TABLE 65–1

Immunologic Aspects of Hypersensitivity Reactions

Type	Sensitization	Immunologic Reaction
I (Immediate, anaphylactic) IgE-mediated	Antigen (allergen) induces IgE antibody that binds to mast cells and basophils.	When exposed to the allergen again, the allergen cross-links the bound IgE on those cells. This causes degranulation and release of mediators (e.g., histamine).
II (Cytotoxic) IgG-mediated	Antigens on a cell surface elicit Tfh and B-cell activation.	Antibody binding to cell membrane antigens leads to complement-mediated lysis of the cells (e.g., transfusion or Rh reactions) or autoimmune hemolytic anemia.
III (Immune complex) Multiple types of antibodies	Soluble antigens elicit Tfh and B-cell activation.	Antigen–antibody immune complexes are deposited in tissues and neutrophils are attracted to the site. They release lysosomal enzymes, causing tissue damage. In the case of IgM or IgG, complement is also involved.
IV (Delayed) T-cell–mediated	CD4 and/or CD8 T cells sensitized by protein antigens.	Memory T cells release cytokines upon second contact with the same antigen. The lymphokines induce inflammation and activate macrophages, which, in turn, release various inflammatory mediators.

Tfh = T follicular helper cell.

Type II Hypersensitivity - Cytotoxic

- 1. Ag on cell surface,
- 2. Attracts IgM and IgG Ab
- 3. Promotes cell lysis via complement system
- Ex: Transfusion Reactions

Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz: Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, Seventeenth Edition:

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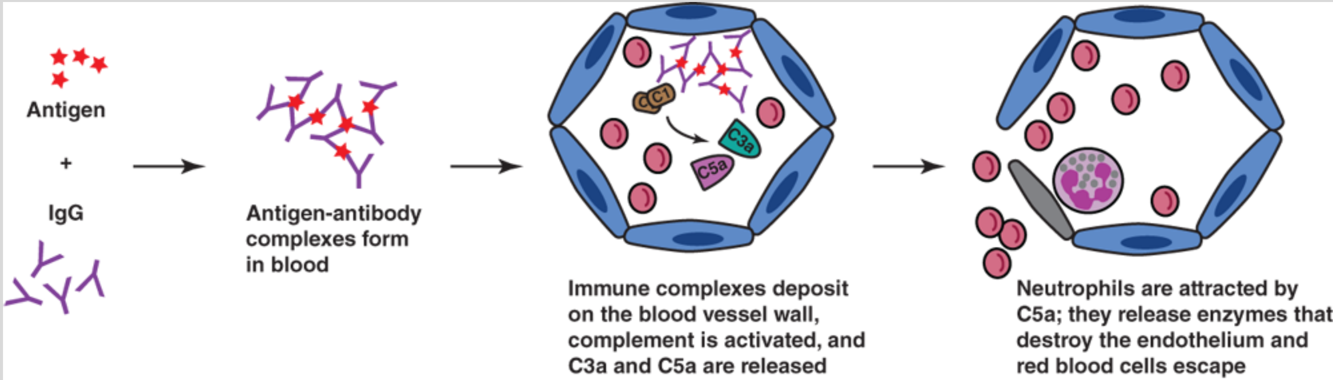
Type III Hypersensitivity - Immune Complex

- Antigen (freely circulating, not on Cell Wall) bind to Antibody
- Deposited in vessel walls
- Causes – Inflammatory Reaction
- Ex: Serum Sickness



Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz: Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, Seventeenth Edition:

Type IV Hypersensitivity–Cell Mediated

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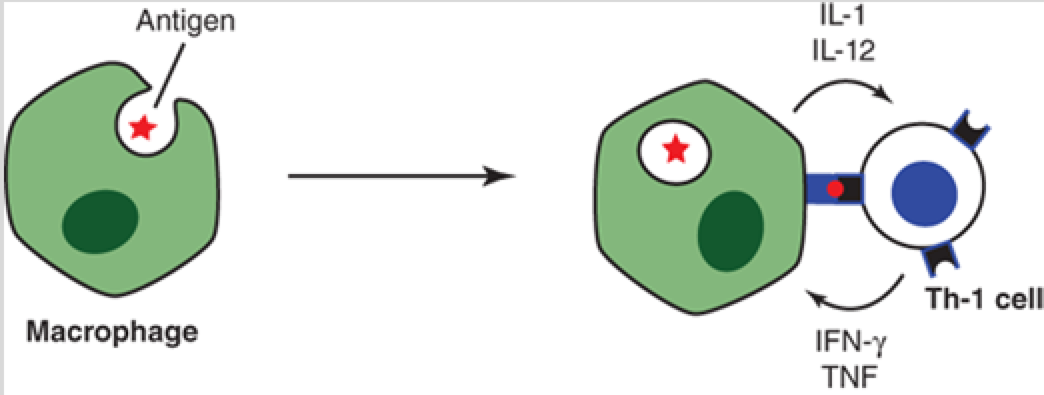
- Previously sensitized T1 helper
- Cells recognize an antigen (Ag Presenting cell) and release cytokines to recruit more T1 helper cells and mononuclear cells
- Inflammatory reaction
- Ex: Contact Dermatitis



Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz: Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, Seventeenth Edition:

Type 1Hypersensitivity

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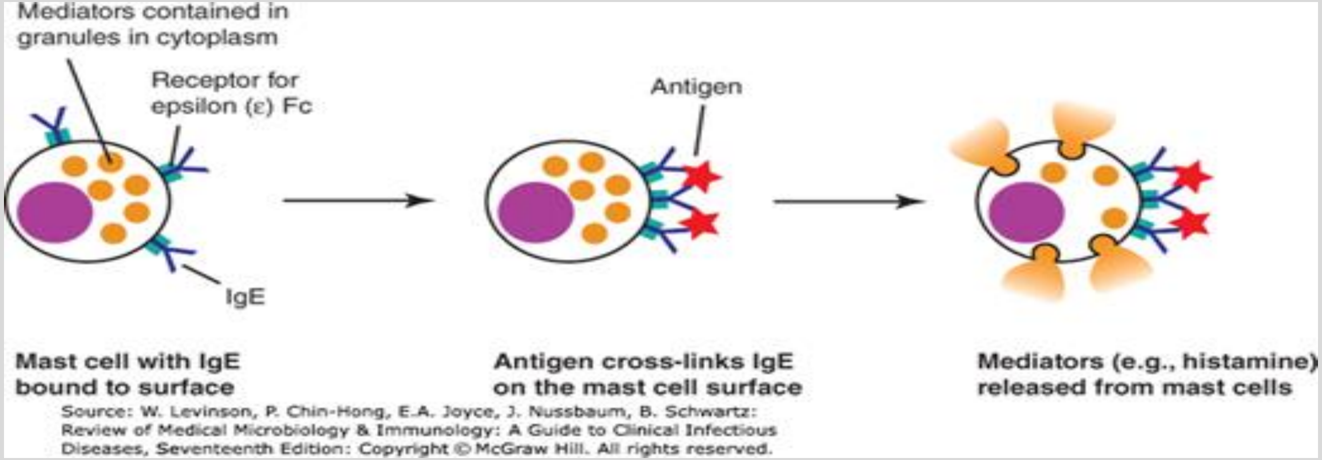
- Two Phases:
- 1. Immediate – allergies/anaphylaxis
- Antigen crosslinks preformed IgE on presensitized mast cells -> degranulation-> release of pre-formed histamine, tryptase, and leukotrienes.



Type I Hypersensitivity

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- Delayed (late hours) –
  - Newly synthesized Leukotrienes ->
  - Attract inflammatory cell (eosinophils)
  - Inflammation/tissue damage
    - Erythema, edema, induration

Mechanism of Type I Hypersensitivity Reaction

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- 1. Exposure to allergen
- 2. IgE produced
- 3. IgE binds to surface of mast cells & basophils
- 4. 2nd exposure to allergen
- 4. Allergen binds to cell associated IgE & cross linking of two IgE Fc receptor molecules on mast cell surfaces.
- 5. Signal transduction in mast cells and basophils
- 6. Mediator secretion (ex: histamine secretion)
- 7. Mediator effects on end organs like blood vessels & bronchial smooth muscles.

Late Phase

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- About 6 hours after exposure
- Due to mediators (leukotrienes) synthesized after cell degranulates
- Influx of inflammatory cells (neutrophils and eosinophils)

- Biphasic Reaction – reaction meeting anaphylactic criteria within 48 hours of previous reaction without re-exposure to Ag

EXPOSURE

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- Primary exposure is Respiratory, GI, and Skin

What cells live in skin, mucous membranes, and connective tissue? Mast cells

Some genetic predisposition to allergies.

Common triggers

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- Foods
- Medications
- Insect Bites
- Immunotherapy

injections

Symptoms

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- Skin – Flushing, Pruritis, Urticaria (hives)
- Respiratory – Rhinitis, Stridor, Wheezing
- GI – Nausea, Vomiting, Diarrhea
- ENT – Uvular edema, Tongue Swelling, Oral pharyngeal edema
- Neuro – Headache, Seizure
- CV – Chest Pain, Hypotension

Source: Kevin J. Knoop, Lawrence B. Stack, Alan B. Storrow, R. Jason Thurman: The Atlas of Emergency Medicine, 5e

Table 6. Differential Diagnosis for Allergic Reactions

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- **Life-Threatening Causes**
- Angioedema
- Acute coronary syndromes
- Arrhythmia
- Airway obstruction or foreign body
- Asthma and other atopic diseases
- Sepsis and septic shock
- Pneumothorax
- Scombroid poisoning
- **Non–Life-Threatening Causes**
- Anxiety
- Panic attack
- Vasovagal reaction/syncope
- Dermatitis
- Gastroenteritis
- Chronic urticaria
- Mastocytosis or other mast cell disorders
- Liver dysfunction

Zeke A. Sudhir A. Management of Allergic Reactions and Anaphylaxis in the Emergency Department. *Emerg. Med Practice* 2022, 24 (7): 1-24.

Diagnosis of **Anaphylaxis**

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- 1. Acute onset of mucocutaneous

Symptoms (flushing, pruritis, urticaria, angioedema)

plus one of the following

  - Wheezing, stridor, hypoxia
  - Hypotension or end-organ damage
- 2. Two or more of the following that occur rapidly after exposure to known or suspected allergen
  - Mucocutaneous Involvement
  - Respiratory Symptoms
  - Hypotension or end-organ hypoperfusion
  - GI Symptoms (pain, nausea, vomiting)

Diagnosis of Anaphylaxis

- 3. Hypotension within minutes to hours after exposure to a known allergen

  - Hypotension = Anaphylactic Shock

Treatment

What do you want to do for your Pt in the picture below after she ate a kiwi?

Source: B.R. Shah, P. Mahajan, J. Amodio, M. Lucchesi
Atlas of Pediatric Emergency Medicine, Third Edition

Treatment

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- Airway Protection
- Epinephrine –
  - 0.3-0.5 mg IM (in the thigh)
  - Peds: 0.01 mg/kg IM
  - Only evidence proven medication for anaphylaxis
    - Increases Peripheral Vascular Resistance and Cardiac Output
    - Stabilizes mast cells
    - Reverses Bronchoconstriction and mucosal edema

Treatment

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- Refractory to IM Epi – use IV
- Decontamination – Remove triggers
  - Ex: insect stinger
- Albuterol if wheezing only – alleviate bronchospasm
- IV crystalloid infusion if hypotensive
- Corticosteroids (evidence low) – anti-inflammatory and may help prevent biphasic reactions

Treatment

- The Anti-histamines:
  - No evidence to support life-threatening aspects of anaphylaxis
  - Helps w/ cutaneous symptoms
  - Is your pt on a Beta-blocker:
    - May need Glucagon

Treatment

- Your pt remains w/ bp of 80/60, wheezing. Pt has been intubated as tongue swelling increased. What else can you do?

- Add additional vasopressors
- Call ECMO team
- Some data on methylene blue use in refractory anaphylaxis

Disposition

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- Low, risk pt, Resolved Anaphylaxis
  - Observe and discharge

  if remain stable
  - Rx for epinephrine auto-injector
  - Consider steroids or anti-histamine
  - Rx for symptoms relief
  - Follow up w/ primary care physician or allergist

Disposition

- High Risk, Mild to Moderate

Anaphylaxis

- Persistent symptoms, increased risk of death from anaphylaxis (Congestive Heart Failure, Chronic Kidney Disease, asthma, hx of severe allergic reactions, pts on Beta Blockers)
- Prolonged ED Observation or Admit

Disposition

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- Severe Anaphylaxis
  - ICU Admission
    - Intubated
    - IV epi pts
    - Airway Monitoring

Non-anaphylactic Allergic Reactions

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- 90% of pts have urticaria
  - Ask about potential exposures
  - New meds? Soaps? Foods? Exposures? Family hx of allergic reactions?

- Not all rashes – allergic reactions
  - Ex: virus in children

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

Non-anaphylactic allergic reactions

- Tx:
  - Antihistamines – symptoms
    - H1/H2 Blockers (cutaneous symptoms)
  - Monitor for progression to anaphylaxis
  - + / - Corticosteroids
    - Decrease itch
    - Risk versus Benefit in mild/moderate reactions
    - Severe – Give Steroids

Summary Slide

- Allergies/Anaphylactic Reactions are Type 1 Hypersensitivity Reactions
- Epinephrine is the life-saving medication for anaphylaxis
- Avoid re-exposure to known allergens when possible

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